

NAVEEN BABU KRISHNASAMY JEYAKUMAR

naveenbabu599@gmail.com

[linkedin.com/naveen-babu](https://www.linkedin.com/in/naveen-babu)

github.com/naveenbabu

+19796351744

Education

Texas A&M University, Texas, USA

Aug 2024 – Dec 2025

Master of Science in Computer Engineering

PSG College of Technology, Coimbatore, India

Jul 2016 – Apr 2021

Bachelor of Engineering in Electrical and Electronics Engineering

Relevant Courses: Operating Systems, Parallel Computing, Micro Processor System Design, Data Structures & Algorithms, Computer Architecture, Microprocessors & Microcontrollers, Computer Networks, GPU Programming and Visualization, Distributed Systems and Cloud Computing

Work Experience

Summer Intern | Qualcomm Inc. | Santa Clara, USA

May 2025 – Aug 2025

- Developed a test framework for MAC Firmware (Wi-Fi Data Path) to support unit and simulation testing and integrated the framework into the CI check-in pipeline.
- Optimized Wi-Fi stats and packet logs for efficient crash-time diagnostics and reduced Wi-Fi stats reporting latency by 40µs per iteration.

Firmware Engineer | Qualcomm Inc. | Chennai, India

Nov 2021 – Jul 2024

- Implemented rate adaptation algorithms for Wi-Fi 6/7 on embedded platforms supporting MU-MIMO and SU TxBF, increasing data throughput by 50 Mbps and improving connection stability.
- Developed Multi-Link Operation (MLO) features such as MLSR and MLMR for SU TxBF and MU-MIMO, enabling robust multi-band communication across 2G/5G/6G links.
- Designed and integrated beamforming features for drone applications, improving real-time control responsiveness and extending operational range to approximately 12 km.
- Developed object-oriented features to enhance WLAN firmware for AP/Gateway products, managed multiple connectivity protocols, and utilized the OpenWrt Linux distribution to operate Linux device drivers.
- Collaborated with customer teams and OEMs to diagnose and resolve critical stability, WFA certification and crash issues in Wi-Fi firmware.
- Leveraged embedded processors and RTOS concepts to enhance system performance that involved reverse-engineering target memory dumps to identify and resolve low-level stability issues.
- Conducted functional, regression, and performance testing on QCA chipsets using DUTs, emulators, and tools like Litepoint for stability and throughput validation.
- Led cross-functional projects across the development lifecycle using Agile methodologies and tools such as Perforce, Git, JIRA, and Jenkins.

Associate Software Engineer | Robert Bosch (BGSW) | Coimbatore, India

Jul 2021 – Nov 2021

- Worked on model-based software development for engine management, powertrain, and vehicle functions for premium automotive OEMs. Tools used: ASCET, Labcar, MISRA C, MATLAB, ETAS INCA.

Student Intern | EMF Innovations Private Limited | Coimbatore, India

May 2019 – Jul 2019

- Designed and developed a driver circuit for SRM motors using PIC micro-controllers, along with a current limiter and an AC to DC controlled converter for electric scooter applications.

Technical Skills

Languages: C, C++, Assembly, CUDA C, Python, MATLAB, Verilog, HTML/CSS.

Tools & Frameworks: Trace32, Linux Kernel, Git, Perforce, MATLAB, QEMU, OpenMP, GDB, tcpdump, POSIX, Jenkins, x86, WFA, JIRA, VSCode, ASCET, Wireshark, Veriwave, IxChariot.

Expertise: Firmware development, RTOS, Low-level programming, Bare metal, Operating Systems (Threads, Process Management, Interrupts, Memory Management, Concurrency, Scheduling, Virtual Memory, Virtualization), Wireless Communication Protocols, Multithreaded Programming.

Protocols: IEEE 802.11, TCP/IP, UDP, I2C, UART, SPI.

Projects

x86-based Mini-Kernel

Aug 2024 - Dec 2024

Developed a mini-kernel for x86 architecture using C++ and tested it using the Bochs emulator.

- Implemented virtual memory management with bitmap based frame allocation and page tables aligned to the x86 MMU.
- Developed thread management including context switching, and implemented both FIFO and round-robin schedulers using timer interrupts.